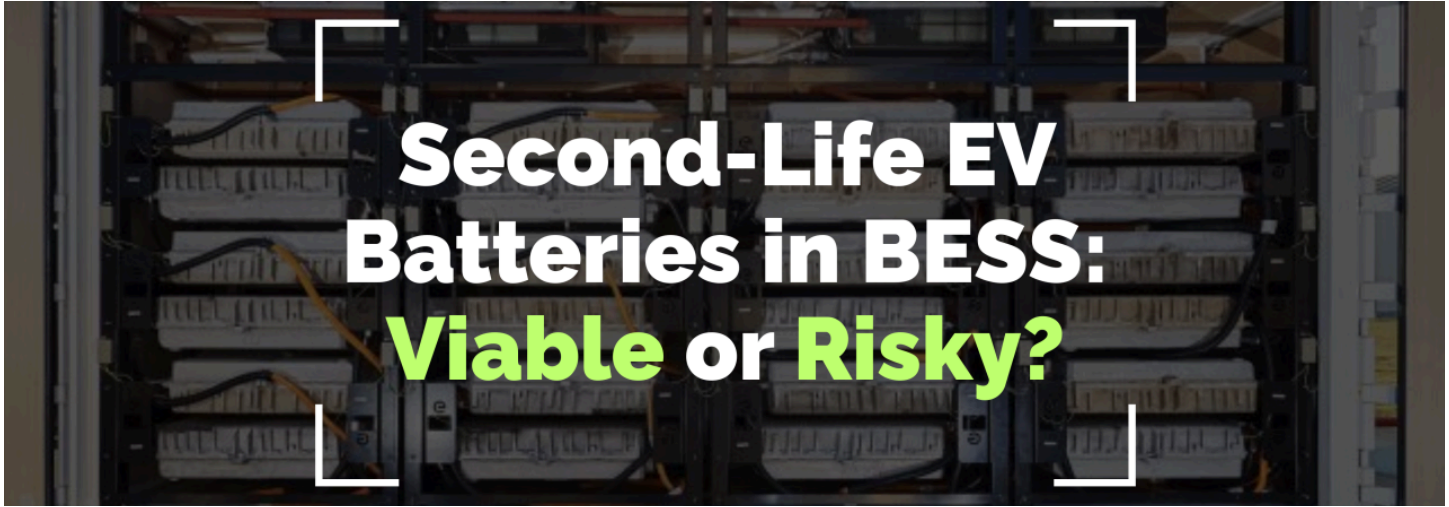




Second-Life EV Batteries in BESS: Viable or Risky?

Second-Life EV Batteries in BESS: Viable or Risky?



Neeraj Kumar Singal

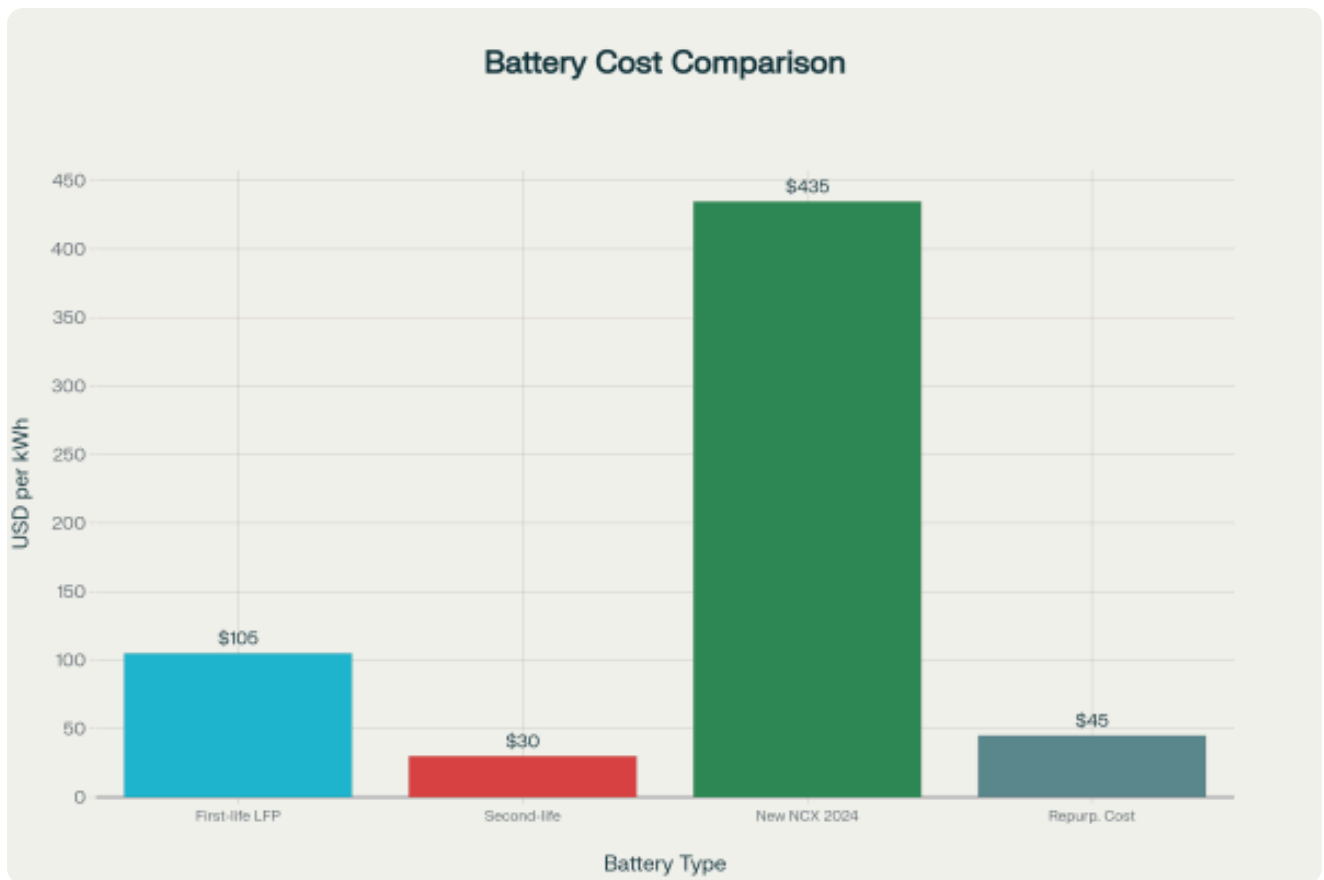
28,602 followers

Published Jul 22, 2025

For grid operators, developers and investors, the idea of repurposing the flood of retired electric-vehicle (EV) batteries into stationary battery-energy-storage systems (BESS) promises lower capital costs and faster progress toward decarbonization. Yet every reused pack also brings uncertainties—from degraded performance to heightened fire risk. This newsletter unpacks the opportunity and the pitfalls in depth.

Executive Summary

Second-life BESS projects are scaling rapidly because repurposed packs can cost 20-35% less upfront than new lithium-ion systems while avoiding 17% of embedded carbon versus direct recycling. Analysts project market values ranging from \$4.7 billion in 2030 to \$28.17 billion by 2031, implying 25-45% compound growth. However, levelized-cost-of-storage (LCOS) estimates remain 234-278 \$/MWh—still above new-build systems—unless discount rates, repurposing costs and cycling depth are optimized. Fire incidents in Korea, Arizona and the UK demonstrate that degraded cells pose real hazards if thermal-runaway barriers, UL 1974 grading, NFPA 855 spacing and robust ventilation are omitted. The bottom line: second-life BESS is viable but only under stringent safety, warranty and financing frameworks.



Market Drivers and Demand Signals

Volume of Retired Batteries

- Global EV stock will exceed 240 million by 2030, translating into 1 TWh of end-of-first-life batteries within a decade.
- Most packs still hold 70-80% capacity when removed from vehicles, sufficient for less demanding stationary roles.

Cost & Carbon Advantages

- Repurposed modules reduce upfront costs to 64.3-78.9% of new-build equivalents.
- Life-cycle analyses show an additional 58% profit and 18% emission reduction for LFP chemistry when reuse precedes recycling.
- Extending life delays raw-material demand at a time when lithium, cobalt and nickel could rise 500% by 2050.

Circular-Economy Policy Tailwinds

- The EU Battery Regulation 2023/1542 mandates digital passports, carbon-footprint labels and second-life access to pack data starting 2027.
- India's Battery Waste Management Rules impose extended producer responsibility for reuse or recycling.

Economics: Crunching the Numbers

CAPEX & OPEX Profile: Repurposed systems avoid cell procurement costs but incur additional testing, disassembly and logistics expenses.

Element	New BESS Share	Second-Life Delta	Notes
Cell/Module Purchase	55%	-35% to -45%	Pack acquired at scrap value
Repurposing & Testing	0%	+10% to +18%	UL 1974 gateway, capacity grading
Balance-of-System (PCS, EMS, HVAC)	30%	0%	Identical hardware
Warranty & Insurance	5%	+3% to +5%	Higher perceived risk
Decommissioning Reserve	1%	+2%	Accelerated EoL uncertainty
O&M (Annual)	3% of CAPEX	+1%	Extra cell monitoring

Cost Savings Cost Increase No Change

Levelized Cost of Storage: Steckel’s harmonized LCOS model yields 234-278 \$/MWh for 15-year second-life projects vs 211 \$/MWh for new packs, heavily influenced by discount rate, depth of discharge and repurposing cost. At discount rates below 6%, second-life parity becomes achievable.

Market Size Forecasts

SOURCE	YEAR OF ESTIMATE	2025 VALUE	2030 VALUE	2031 + VALUE	CAGR
	2024	\$1.2 billion	\$4.7 billion	N/A	25.5%
	2023	\$0.53 billion	\$9.93 billion (2031)	\$9.93 billion	44.7%
	2024	N/A	N/A	\$28.17 billion (2031)	43.9%
	2024	N/A	N/A	\$4.2 billion (2035)	22% est.
	2025	\$28.52 billion	\$47.46 billion (2030)	N/A	10.6%

Financing, Insurance and Warranty

- Data-driven performance guarantees from Altium/Tokio Marine Kiln lower lender risk premiums.
- Insurance underwriters increasingly demand early design engagement, minimum spacing and detailed degradation models to price property and liability coverages.
- Developers must budget higher deductibles or sub-limits for projects using mixed-origin packs.

Technical Performance Barriers

State-of-Health (SoH) Variability: Retired packs exhibit heterogenous capacities (60-85% SoH) and internal resistance profiles, complicating module balancing.

Battery Management System (BMS) Retrofitting: OEM BMS firmware may hinder access; open-protocol retrofits or “middleware BMS” layers are emerging to aggregate packs from multiple brands while respecting CAN bus security.

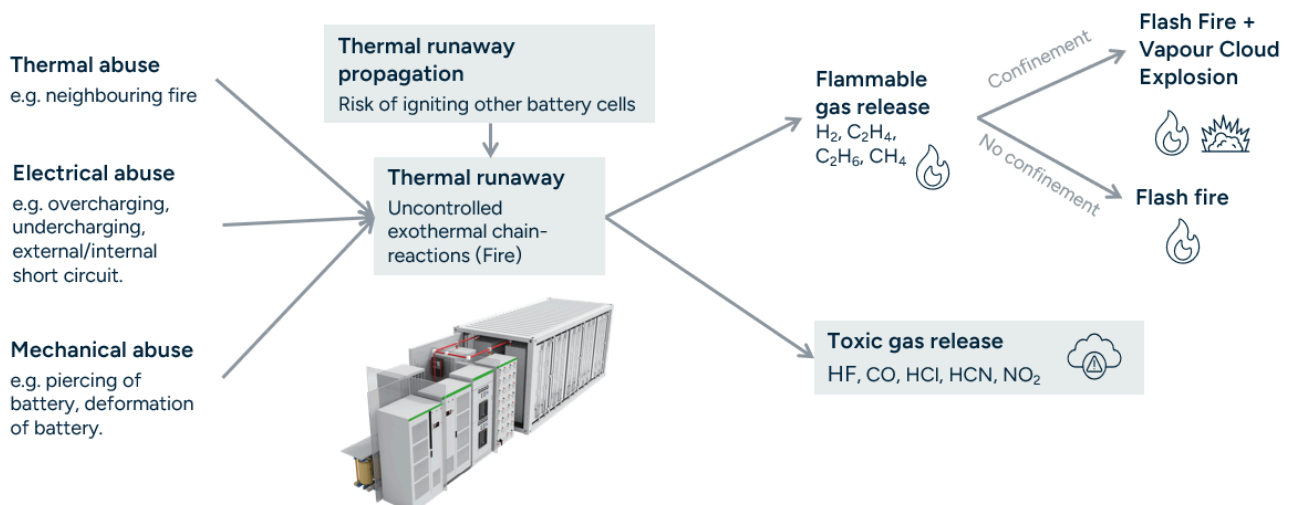
Cycle Life Expectations: While EV duty cycles degrade cells quickly at high C-rates and extreme temperatures, stationary applications with 0.2-0.5 C operation extend life by 8-10 years—about 3,000 additional cycles.

Integration Architecture: Containerized pods or rack-based retrofits must accommodate varying form factors (pouch, prismatic, cylindrical). Custom busbar and cooling designs add engineering time but allow mixing chemistries in separate strings under a unified EMS.

Safety and Risk Profile

Thermal Runaway Incidents

- 30+ large-scale BESS fires recorded globally since 2018, with many in second-life systems lacking robust ventilation.
- Arizona 2019: 2 MWh system explosion injured firefighters; root cause traced to cascading failure from one cell with latent defect.



Hazard Pathway: Four stages: abuse factor → off-gassing (CO, H₂) → smoke → flame. Off-gas sensors such as Li-ion Tamer provide early detection before smoke phase.

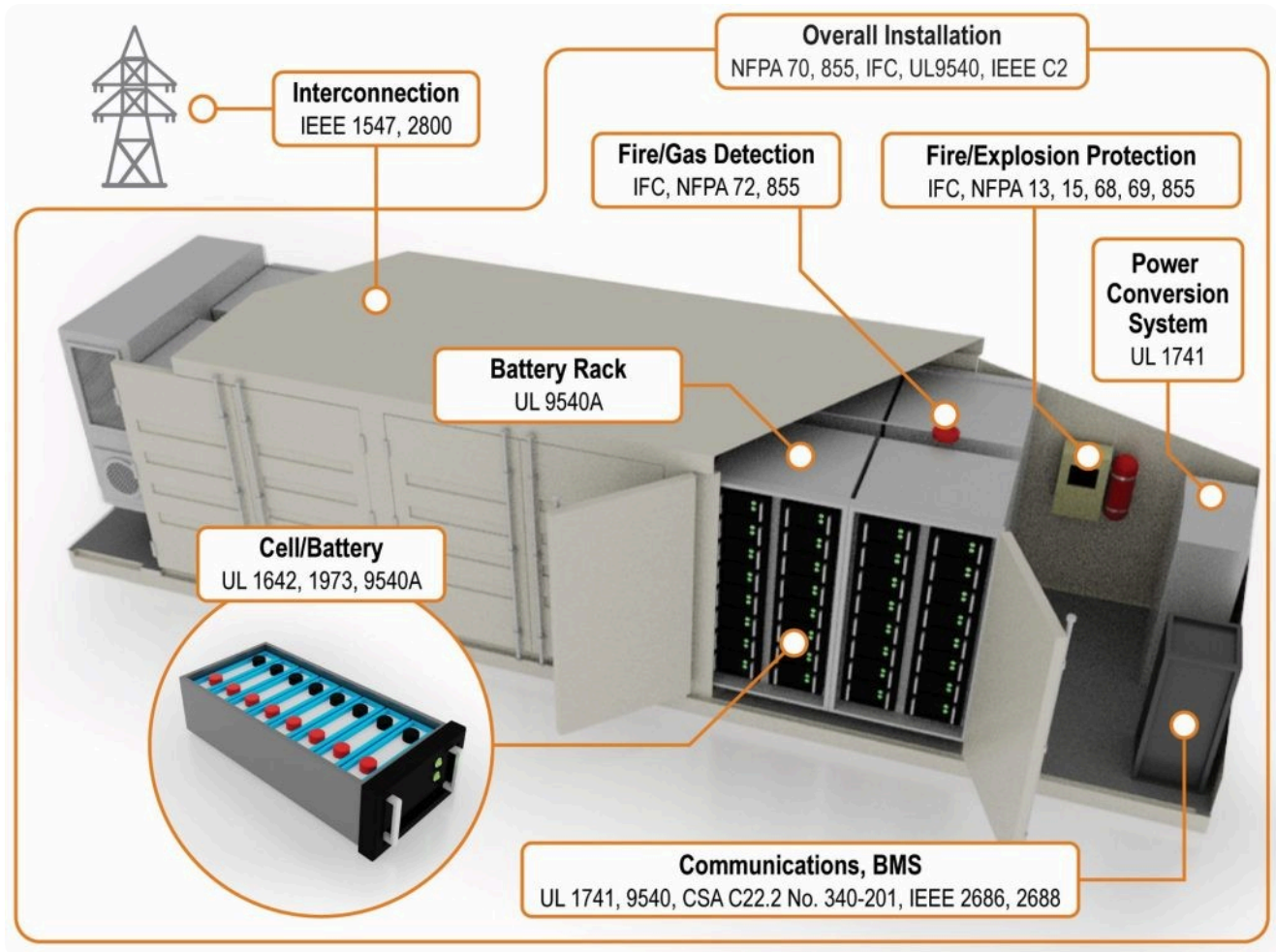
Codes, Standards and Best Practice

STANDARD / CODE	STATUS	SCOPE	KEY REUSE PROVISIONS
UL UL 1974	ACTIVE	Sorting & grading of repurposed packs	▶ Expiration-date check ▶ SoH gating ▶ Routine maintenance tests
UL UL 9540	ACTIVE	System-level safety	▶ Thermal runaway mitigation ▶ BMS failsafe ▶ Fire-resistance rating
NFPA NFPA 855	ACTIVE	Installation	▶ Unit spacing ▶ Ventilation ▶ Deflagration panels
IEC IEC 63330	DRAFT	Second-life evaluation	▶ Global harmonization in progress
EU EU Reg. 2023/1542	ACTIVE	Lifecycle & passport	▶ Mandatory data access for reuse ▶ QR label for SoH

Mitigation Techniques

- Cell-level fuses and module-level fire barriers to halt propagation.
- Active cooling redundancy; liquid cooling preferred for high-density racks.
- Gas detection linked to automated HVAC exhaust.

- Emergency response plans rehearsed with local fire services; water-mist and aerosol systems validated for Li-ion suppression.



Business Models and Real-World Deployments

Connected Energy (UK/EU): Fleet of 1st-generation Renault-Nissan modules powering 300 kW to multi-MW systems since 2010; field data confirm 10-year secondary life expectancy.

JLR + Allie Portable BESS: Seven PHEV packs deliver 270 kWh per trailer-mounted unit for event power; directly reuses OEM BMS without cell teardown.

Einride Smartcharger Stations: Decommissioned truck packs store PV energy to enable low-cost overnight charging; digital platform predicts degradation and trading value.

GM Milford Data Center: Chevrolet Volt packs buffer onsite solar-wind microgrid, demonstrating modular reuse with minimal remanufacturing.

Environmental and Lifecycle Assessment

17-18% GHG Reduction

58% Profit Boost

70kg Lithium Saved

Second-life use shifts environmental burdens:

Scenario	GHG Saving vs Immediate Recycling	Resource Conservation
1x Reuse Then Recycling	17% lower cradle-to-grave CO _{2e}	Delays virgin lithium demand by 70 kg per 40 kWh pack
LFP Pack Reuse in ESS	18% emission cut, 58% profit boost	Superior vs NMC due to cobalt-free chemistry
Swedish Residential PV + 2nd-life BESS	Light NPV advantage in multi-family houses at ≥800 cycles	Sub-optimal under current tariff; improves with real-time pricing

Viability Matrix by Use Case

Application	Revenue Model	Technical Stress	Fire Risk	Net Viability
Front-of-meter Frequency Regulation	Capacity payments & arbitrage	Moderate cycling	Medium (high C-rate)	Medium
C&I PV Self-consumption	Demand charge reduction	Low cycling	Low	High
Rural EV Fast-charging Buffer	High-power peaks	High C-rate	Medium	Medium
Backup/UPS	Stand-by float	Minimal cycling	Low	Very High
Mobile Event Power (JLR)	Rental fees	Intermittent	Low	High
Microgrid Islanding	Daily deep cycles	High	Medium-High	Medium

Risk Management Toolkit

Technical Due Diligence Checklist

1. Verify UL 1974 certification of supplier and lot traceability.
2. Conduct 100% OCV, IR and SoH testing; reject packs <70% capacity.

- 3. Simulate string-level degradation impact on round-trip efficiency.
- 4. Require UL 9540A test report for container design and ventilation.

Insurance Best Practices

- Engage insurers pre-design to align on clearance and fire-wall schemes.
- Detail cyber-security hardening for remote BMS control to qualify for cyber extensions.
- Negotiate contingent-business-interruption cover tied to EPC delays.

Warranty Structures

- Performance-linked premium models (Altelium/TMK) adjust yearly based on real BMS data, freeing OEM capital.
- Minimum warranty: 70% capacity retention for 7-10 years or specific MWh throughput.

Regulatory Watchlist

Region	Current Status	Upcoming Milestones
 European Union	Regulation 2023/1542 in force; digital passport by 2027	Reuse gateway conformity assessment list by 2026 
 United States	UL 1974 voluntary; NFPA 855 adoption varies by state	Proposed federal "Battery Stewardship Act" debate 2025 
 United Kingdom	OPSS second-life BESS safety report guiding future standards	Possible mandatory gateway testing 2026 
 India	Battery Waste Management Rules enforce EPR	Draft reuse guidelines expected 2025 
 China	MIIT pilot programs trade retired packs via data platform	National second-life standard GB/T update 2026 

Strategic Recommendations

Developers & EPCs

- Prioritize applications with low C-rate duty cycles and stable revenue (backup, peak-shaving).

- Model LCOS across degradation scenarios; include repurposing capex sensitivity.
- Secure data-driven performance warranties to unlock cheaper debt.

Investors & Lenders

- Demand third-party SoH audit and UL 1974 compliance before financial close.
- Price higher contingency for mixed-chemistry systems; favor homogeneous NMC- or LFP-only strings.

Policymakers

- Provide investment tax credits parity for second-life storage to level LCOS gap with new batteries.
- Mandate open BMS data sharing to stimulate grading market and prevent “data lock-in”.

OEMs

- Design packs for easier module replacement, embedded QR battery passport, and lower disassembly cost.
- Monetize residual value via take-back partnerships, turning EoL liability into revenue.

Outlook to 2035: Scenarios

SCENARIO	2035 INSTALLED CAPACITY	KEY DRIVERS
Conservative	110 GWh	• Slow policy uptake across markets • Limited data access and transparency
Base Case	205 GWh	• EU & US adopt digital passport systems • LCOS parity achieved at \$200/MWh
Accelerated Circularity	320 GWh	• Global carbon pricing exceeds \$100/tonne • OEM buy-back mandates implemented

Even in the conservative path, second-life batteries could supply nearly one-quarter of projected stationary-storage demand, easing raw-materials pressure and cutting system costs.

Conclusion

Second-life EV batteries are not a universal remedy, but where policy, engineering and financing co-align, they offer a pragmatic bridge to deeper decarbonization. The technology is **viable**—provided rigorous safety standards, transparent data and robust warranties are in place. Ignore those guardrails, and the approach quickly becomes **risky**. The next decade will reveal which players master the balance.

For More Updates Follow Us

[WhatsApp](#) - [Facebook](#) - [Instagram](#) - [Twitter](#) - [LinkedIn](#) - [YouTube](#)